

Phase 1: DevOps & CI/CD (Jenkins Focus)

1. What is the difference between Continuous Integration, Delivery, and Deployment?
2. How does Jenkins integrate with Git?
3. What is a Jenkins Pipeline and why use it over Freestyle projects?
4. Explain the concept of "Pipeline as Code."
5. What are Jenkins Agents (Nodes) and why are they used?
6. How do you handle a broken build in Jenkins?
7. What is a Jenkinsfile?
8. Explain the difference between sh and bat steps in Jenkins.
9. How do you backup or move your Jenkins master?
10. What are "Stages" and "Steps" in a Declarative pipeline?
11. How do you use Environment Variables in Jenkins?
12. What is a Webhook?
13. How do you secure Jenkins (User Management/RBAC)?
14. What is the purpose of the "Post" section in a Jenkins pipeline?
15. How do you handle secret credentials in Jenkins without hardcoding them?

16. What is a Shared Library in Jenkins?
17. Explain the "Poll SCM" feature.
18. How do you trigger a job after another job finishes (Upstream/Downstream)?
19. What is a "Multibranch Pipeline"?
20. How do you clean up the workspace in Jenkins after a build?
21. What is the role of the agent any directive?
22. How do you run parallel stages in Jenkins?
23. What is the difference between Scripted and Declarative pipelines?
24. How do you integrate SonarQube with Jenkins?
25. Explain the importance of Artifact Management (Nexus/JFrog) in CI/CD.

Phase 2: Docker (Containerization)

26. What is the difference between a Virtual Machine and a Container?
27. Describe the Docker Architecture (Client, Host, Registry).
28. What is a Docker Image?
29. What is a Docker Container?

30. Explain the purpose of a Dockerfile.
31. What is the difference between CMD and ENTRYPOINT?
32. What is a Docker Image Layer?
33. How do you reduce the size of a Docker Image?
34. What are Multi-stage builds?
35. How do you persist data in Docker (Volumes vs. Bind Mounts)?
36. What is Docker Hub?
37. Explain docker-compose.yml.
38. How do you view logs of a running container?
39. What is the difference between docker run and docker start?
40. How do you remove all stopped containers?
41. What is Docker Swarm (Briefly)?
42. How does Docker networking work (Bridge, Host, None)?
43. What is the .dockerignore file?
44. How do you push an image to a private registry?
45. Explain the EXPOSE instruction in a Dockerfile.

- 46. What is the difference between COPY and ADD?
- 47. How do you check the resource usage of containers?
- 48. Can a container stop by itself? Why?
- 49. How do you execute a command inside a running container?
- 50. What is an orphaned volume in Docker?

Phase 3: Kubernetes (Orchestration)

- 51. What is Kubernetes (K8s)?
- 52. Explain the Master-Worker Node architecture.
- 53. What is the role of the API Server?
- 54. What is etcd?
- 55. What does the Scheduler do?
- 56. Explain the Controller Manager.
- 57. What is a Pod?
- 58. Why can't we run containers directly in K8s instead of Pods?
- 59. What is kubectl?

60. What is a Deployment in K8s?
61. Explain Replicasets.
62. What is a Kubernetes Service?
63. Difference between ClusterIP, NodePort, and LoadBalancer services.
64. What is an Ingress Controller?
65. Explain ConfigMaps and Secrets.
66. What are Liveness and Readiness Probes?
67. What is a Namespace?
68. How do you scale a deployment in K8s?
69. What is a Persistent Volume (PV) and Persistent Volume Claim (PVC)?
70. Explain the role of the Kubelet.
71. What is Kube-proxy?
72. What is a DaemonSet?
73. What is a StatefulSet?
74. How do you perform a Rolling Update in Kubernetes?
75. How do you Rollback a deployment?

- 76. What is a "Headless Service"?
- 77. Explain Resource Quotas and Limits.
- 78. What is HPA (Horizontal Pod Autoscaler)?
- 79. What is a Sidecar container pattern?
- 80. How do you troubleshoot a Pod in Pending state?
- 81. What is a Taint and Toleration?
- 82. What is Node Affinity?
- 83. Explain Helm and why it is used.
- 84. What is the difference between `kubectl apply` and `kubectl create`?
- 85. How do you see the events of a specific namespace?

Phase 4: Advanced Scenarios & Troubleshooting

- 86. How do you automate the deployment of a Dockerized Laravel app using Jenkins?
- 87. What is GitOps (e.g., ArgoCD)?
- 88. How do you monitor K8s clusters (Prometheus/Grafana)?
- 89. Explain "Blue-Green" vs "Canary" deployment.

90. How do you handle database migrations in a CI/CD pipeline?
91. What is "Self-healing" in Kubernetes?
92. How do you secure a CI/CD pipeline (DevSecOps)?
93. What is a "Static Code Analysis" tool?
94. How do you manage infrastructure as code (Terraform vs Ansible)?
95. What happens if the Master Node goes down in K8s?
96. How do you optimize a Jenkins pipeline for speed?
97. Explain the "Zero Downtime" deployment concept.
98. How do you handle logs in a distributed container environment (ELK Stack)?
99. What is a "Build Cache" and how does it help in Docker/Jenkins?
100. Describe your ideal DevOps workflow from code commit to production.